A Framework for Smart Retailing and NFC Mobile Payment Services

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Abstract

Internet of things is an emerging area in the field of Information Systems and Artificial Intelligence. This paper is attempting to integrate internet of things with the virtual mobile payment services via near field communication. In retail purchasing, IoT could be of a great way of minimizing the time taken at the billing counter as well as for the payment. Therefore this study provides a framework where IoT works in coordination with NFC to provide a set of smart device which would help the retailers manage billing and payment. It can also be extended to the inventory management and to have a record of the products being sold based on location and, frame strategies and promotional activities based on sales in each group. The contributions of the study help managers to develop strategies, perform market segmentation, enable e-shopping cart, less consumption of time, less manpower requirement.

Introduction

Innovations in communication mediums, internet, and information technologies in combination with sensors had disrupted the digital era entirely with the evolution of "Internet of Things." The network of interconnected devices enabled with sensors is becoming the widely used technology to harvest information from the surroundings. These interconnected network devices interact with the physical world via the internet with the help of actuators, command, control, information transfer, data analytics, etc. The promising feature of the internet of things (IoT) is to create a world where objects networked through internet communicate among themselves with the least human involvement, essentially integrating the physical and digital world. To achieve this, the objects need to be promoted to smartobjects that simply understands human requirement. With the prevalence of wireless devices such as radio frequency identification (RFID), Bluetooth, Wi-Fi, near field communication (NFC) etc., IoT has gained reinforcement which would revolutionize the digital world completely. The inventions and smart devices account for 22 billion in 2022 for interconnected devices across the world which is expected to reach 64 billion by 2025 (Gartner, 2021)[1].

According to the Boston Consulting Group (2018), by 2020, companies will be spending around 310 billion USD a year on the Internet of Things, with much of this spent in the manufacturing, energy and transportation industries. According to International Data Corporation, global spend on IoT stood at $772.5 billion in 2018 which is expected to reach $1.1 trillion in 2021. Apart from this, if a provision is created to associate smart objects with the applications on mobile devices, it will further add to the ubiquity of these devices. These would open opportunities not only for mobile network operators as well as in the manufacturing sector, automobiles, healthcare, consumer electronics, security, etc. IoT works on the paradigm of cloud computing which not only stores huge amount of data as well as provides tools to analyze and process these.

Numerous definitions of IoT have been proposed by Kevin Ashton, the cluster of European research projects, Forrester research, RFID group, and Atzori (Gubbi et al., 2013; Perera et al., 2014). As the present study has its ground in retailing, therefore a user-centric approach has been adopted to define IoT. IoT is defined as an interconnection of smart objects with internet and communication medium having the ability to share information across platforms through a unified network while autonomously reacting to the commands with or without human involvement. It would function seamlessly with the help of ubiquitous sensors, data analytics, and cloud computing. IoT can be applied to almost everything where we need intelligent systems or object to understand the human requirement. For example, it is applicable to the production units, scheduling systems in industries, smart home, intelligent thermostats, security systems, smart energy applications such as smart electricity, gas and water meters, vehicle fleet tracking and mobile ticketing, smart healthcare facilities such as patients' surveillance and chronic disease management, smart city, smart delivery system, etc. In the present context, we are attempting to combine smart retailing with a virtual payment method via near field communication (NFC). Therefore, the objective of the study is to provide a framework which would enable retail through IoT devices and further integrating it with NFC payment to ensure smart retailing.

Evolution of IoT:

Framework of IoT (Wortmann and Fluchter, 2015)

IoT is comprised of the internet and things and its implementation require both multiple hardware and software components. IoT has three major components: the thing or device layer, the connectivity layer and the IoT cloud layer. The device layer is composed of hardware (system or device) software (programs or algorithms to run the hardware) and IoT components such as sensors, actuators, and processors which work in coordination among each other. The connectivity layer is responsible in establishing communication between the thing layer and the IoT cloud. Further, the software layer is comprised of four sub layers; communication and management layer, the application platform layer, analytics and data management layer and process management layer, and IoT applications layer (Porter and Heppelmann, 2014). The communication and management layer and application platform layer works on setting up software connectivity and provides a platform for IoT implementation. The main components of IoT cloud are analytics and data management, process management, and IoT application. The former two are used basically for storing the huge amount of data, execute, monitor, and analyze and the latter is used for interaction among devices and human.

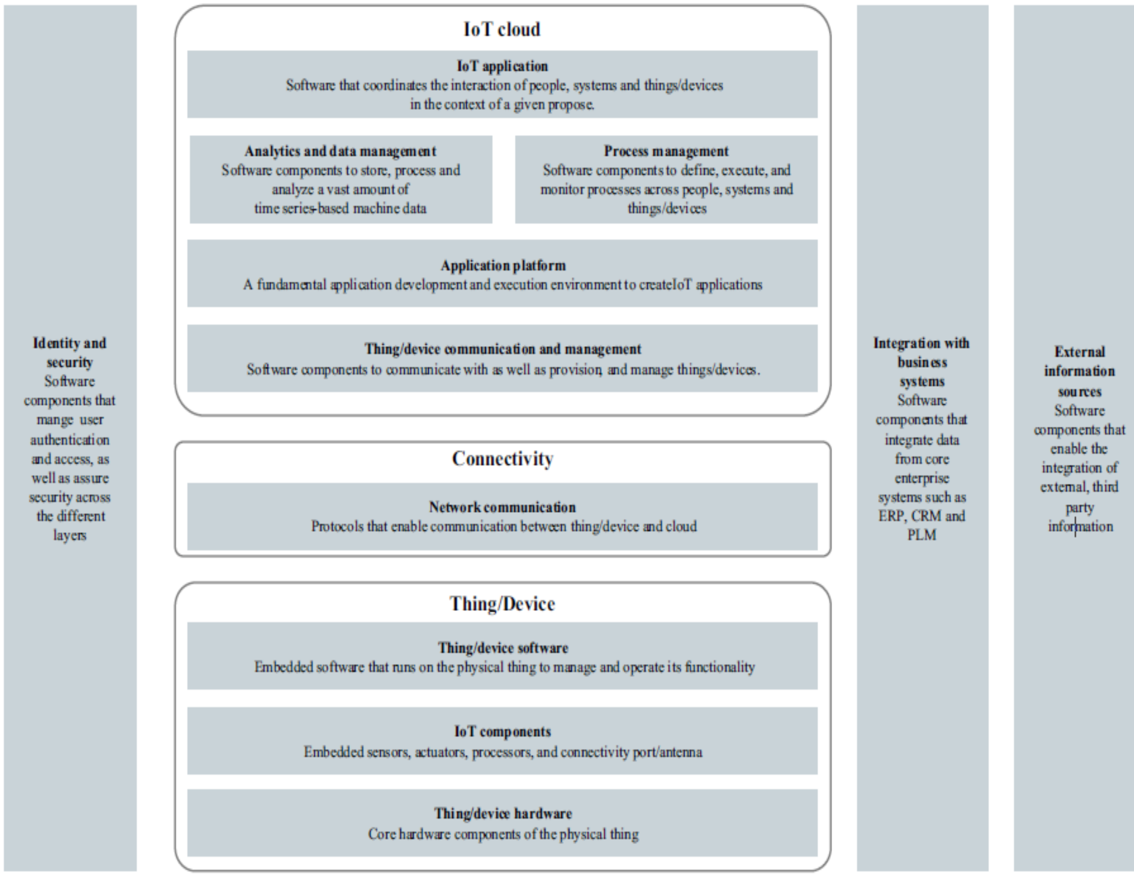


Fig. 1 IoT Framework (Porter and Heppelmann, 2014)

NFC enabled mobile payment services

With the advancements in technologies in various areas, IoT finds its application in the mode of payment too. NFC is a virtual mode of payment where the digital version of our physical card is being stored in our smartphone, with an NFC device installed on the smartphones. The point-of-sale (POS) machines used by the vendors should also have the NFC device installed within, which is connected to the internet. NFC payment occurs when smartphone and POS device is brought in close proximity, where the POS machine reads the virtual card available in the smartphone of the customer. The amount is provided in the POS machines which require one-time-password by the consumers for the successful completion of transaction.

IoT Enabled Technologies

There are several IoT enabled technologies which have emerged as its backbone. Some of them are RFID, wireless sensor networks, Bluetooth, Wi-Fi, NFC, etc. (Kortuem et al., 2009). RFID has been in existence since long specifically in the retail and supply chain which enables wireless data communication via microchips. It works with the help of an electronic barcode which is unique and through which the reader communicates with the product. This automatic identification uses the power of readers' signal for communication and it's usable only until the product reaches its destination. However, the cost of RFID tags are very high, therefore, NFC tag which is cheaper in comparison to RFID could be of great use due of its ubiquitous functionalities. According to ABI research (2018), nfc is being readily used in smart phones by consumers as well as nfc tags are affordable and versatile as compared to other QR codes. Nfc tags costs approximately $0.05 per unit if purchased in high volume. IoT uses the concept of NFC tags on the product which can help store, control, and manages information with or without human intervention. This would replace not only RFID but it can be further used by retailers and consumers for various purposes. Consumers will have complete information about the product, validate its authenticity, its manufacture and expiry date, etc; whereas retailers can keep track of stocks, its usage condition, enable fast billing, can keep track on the consumption of product based on location and other criterion, can create store layout, and need less number of personnel in the store. Thus, the benefits of using NFC tags on the products are multifold. This study proposes retailing based on NFC tags to reap its maximum benefits, of which the functionalities are described below.

E cart diagram

A shopping cart is connected with NFC enabled barcode scanner. NFC Tags (Monteiro, Rodrigues, & Lloret, 2012) contain small microchips with small omnidirectional antennas which can store a small amount of data so as to transfer it to another NFC device such as a mobile phone or a NFC based laptop with few meters. Different NFC Tags have different memory capacities according to which it stores the information in a specific data format (NDEF - NFC data exchange format) (Cheng, Chen, Chi, & Chen, 2009) which can be read by most of the devices. Scanned data will be stored in structured format which will be maintained in a Raspberry Pi. Raspberry Pi is a low cost computer system that consumes minimal power (5volt) and connected with Wi-Fi and Bluetooth. Raspberry Pi will send the data to local server.

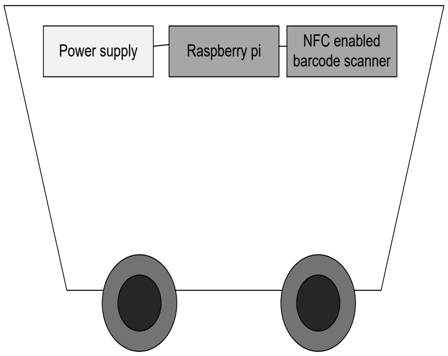


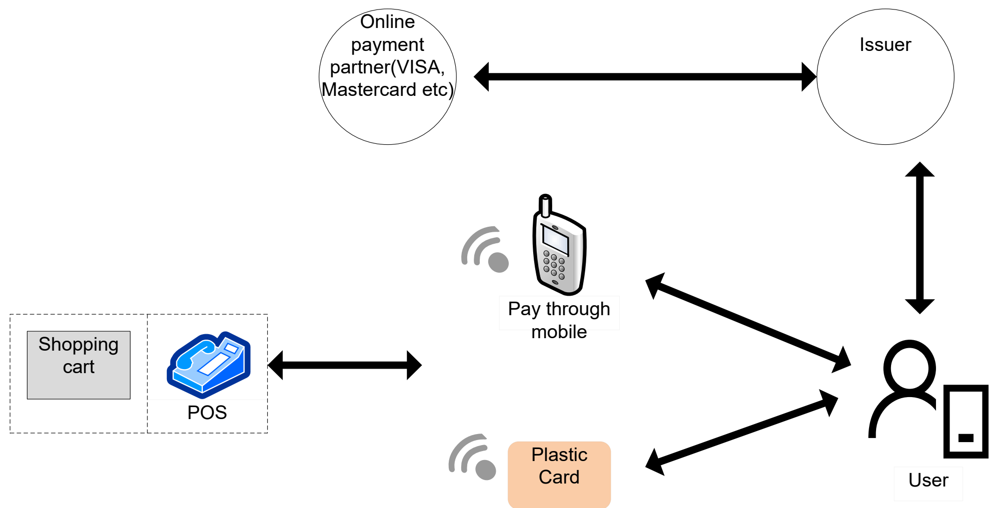
Fig. 2 Flow of smart retail and NFC payment

NFC based e Kart solution:

According to Khan et al. (2017) NFC is a short range wireless connectivity standard (Ecma-340, ISO/IEC 18092) that uses magnetic field induction to enable communication between m-devices when they're brought nearer or touched together. It operates in 13.5 MHz with 424 Kbps speed. The technology used in NFC is based on older RFID (Radio-frequency identification) ideas, which uses electromagnetic induction in order to transmit information between two loop antennas located within each other's 'near field'. It uses an initiator and a target; the initiator actively generates an RF field that can power a passive target.

We proposed a NFC based e cart system that works as follows:

1. Login of customer through android/IOS app on NFC enabled phone.
2. Customer/user walks through the store
3. User enter customer id in the shopping cart.
4. Scan the barcode (attached in the cart) before placing the items in the cart
5. The cart updates item details against customer id in the local server.
6. User can pay the bill either through NFC enabled mobile (apple pay, Google pay etc) or through NFC enabled card during checkout.
7. User can tally the items against payment as the details will be visible in the mobile.



NFC based transaction workflow

Advantages of Smart Retail and NFC Mobile Payment Services

Customers are the backbone of marketing research. Several researchers have attempted to understand customer behavior which is highly differing in different situations. The advent of IoT has provided scope to the marketers to control, manage, and understand the customers' needs and their desires precisely. Moreover, the present study helps the marketers to function the retail industry smoothly and smartly. IoT has several advantages in retail applications.

i. The products can be monitored in real-time, products availability in shelves and inventory.

ii. The expiry date of the products would be helpful in its fast movement and later in its disposal.

iii. Huge database will be created by capturing the data of specific customers with their purchases along with their location.

iv. The identification of physical location in which the consumption of a particular product is high and focusing on other locations with low consumption.

v. The security of the product increases by limiting thefts.

vi. The unique identifier would help in reducing counterfeit products, thus eradicating duplicity.

vii. By generating an electronic bill, the natural resources (papers) can also be preserved.

viii. NFC payment services would require less use of cash transactions.

ix. NFC payment services act as a virtual wallet which reduces the use of physical wallet.

x. The complete transaction would be fast, saves time, money and effort with less requirement of battery backup in the smartphone.

Disadvantages of Smart Retail and NFC Mobile Payment Services

IoT and NFC mobile payment services have a lot of advantages; however, these innovations are never devoid of risks and issues. The disadvantages or the probable issues in IoT and NFC mobile payment services could be:

i. The storage and analysis of huge information collated from these smart devices would be performed through cloud computing. The drawback could be data leakeage as the data are stored in remote cloud data centre which is under the jurisdiction of different data storage policy.

ii. In continuation to the above point, the increase in IoT devices would encounter congestion in network traffic which would further affect the real-time service deterioration.

iii. Since these smart devices would use cloud services, cost will increase.

Contributions and Future research directions

IoT being the emerging area with its ubiquitous advantages is still in its nascent stage. As the internet has undergone enormous changes, so would be the future of IoT. IoT is turning out to be a primary enabler for digital transformation especially large industries. The contributions of IoT have several underpinnings in academics as well as in business. It has opened a pathway for numerous researches on which academicians would be working on; for example, several aspects where IoT can be applied upon, its acceptance among people, how can its maximum benefits be reaped, and its impact on the society. Moreover, this would help managers to frame guidelines for its proper functioning and its implementation issues and the area where it can be applied upon. Apart from this, academicians in collaboration with managers would be able to highlight the areas which might need special attention, to frame different strategies which would have an impact on the performance of several devices, in efficiently managing environmental hazards, etc. Additionally, business objectives such as marketing automation, cost effective, access to sales data, customized services, targeted customer services, market segmentation, improved supply chain, to better understand demand and supply, etc. Retailers can also reduce operating cost, evaluate store footfall by hours, days, time, month, seasons, etc which can further help them assign employees based on requirements. Retailers can also set up help button in mobile app which would further assist customers and resolve their enquiries about the product. Thus, IoT adoption in retail could be a promising field as retailers could leverage their business operations.

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